

Systems Programming

Lab 07 – Shell Scripting

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➤ Lab Objectives

The purpose of this lab is to introduce students to **shell scripting** in Linux by automating the execution of a **TCP client–server guessing game**.

Students will learn how to:

- ✓ **Write and execute shell scripts**
- ✓ Automate program compilation
- ✓ **Run multiple programs automatically**
- ✓ **Execute processes in background**
- ✓ Use delays (sleep)
- ✓ Manage client/server execution order
- ✓ Dealing with **grep command**

Task 1: Automating Client–Server Execution Using Shell Scripting

➤ Problem Description:

Using the previously implemented guessing game **clients.c** and **server.c**:

Write a **shell script** that:

1. Compiles both programs automatically
2. Starts the server first
3. Waits briefly (using sleep) to ensure the server is ready
4. Starts the client automatically
5. Allows both programs to communicate normally

➤ **Sample Output:**

```
Compilation Successful!
=====
Starting Server...
=====

Waiting for server to initialize...
[+] Server socket created.
[+] Bound to port 5000
Listening...

=====
Starting Client...
=====
[+] Client socket created.
[+] Connected to server.
[+] Client Connected.
Enter your guess: Target Number = 91
91
Client guessed: 91

Client won after 1 attempts!
Server Response: Correct
🎉 You guessed correctly!

Stopping Server...
./Task1.sh: line 42: kill: (27089) - No such process

=====
Game Finished
```

➤ **Task 2: Use the grep command in the shell script**

Create the following Employees text file on your machine:

```
Batool AlOmari, HR, 2021  
Ibrahim Alghanem, it, 2023  
Ali Mohammed, Finance, 2020  
Suhaib Al-Obeidallah, IT, 2023  
Lujain Alqudah, hr, 2019  
Suhad Altamimi, HR, 2016
```

Write a shell script that does the following using grep:

- A. Display all lines that start with the letter "S".
- B. Display all lines containing the string "HR" (case-sensitive).
- C. Display all lines containing the string "HR," ignoring case sensitivity.

➤ **Sample Output:**

```
Task 2  
names start with the letter S  
Suhaib Al-Obeidallah, IT, 2023  
Suhad Altamimi, HR, 2016  
names containing HR case senstive  
Batool AlOmari, HR, 2021  
Suhad Altamimi, HR, 2016  
names containing HR case insensitive  
Batool AlOmari, HR, 2021  
Lujain Alqudah, hr, 2019  
Suhad Altamimi, HR, 2016
```